Data-Driven Analysis of University Past Papers and Lecture Content for Curriculum Optimization

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***Abstract*— In modern education, the challenge of effectively optimizing curricula and enhancing student learning through automated systems is becoming increasingly important. This paper presents a novel AI-driven framework aimed at analyzing university past papers and lecture content to facilitate curriculum optimization. The proposed system integrates multiple advanced techniques in artificial intelligence, including Natural Language Processing (NLP), machine learning, and data-driven methodologies. By automating the generation of model exam papers, aligned short notes, personalized study plans, and continuous assessment guidance, the system addresses current gaps in educational tools. It supports the creation of curriculum-aligned, difficulty-controlled exam papers, provides intelligent feedback, and helps identify weak learning areas for targeted improvement. The framework utilizes semantic analysis and retrieval-augmented generation (RAG) techniques to create dynamic, personalized learning experiences that adapt to individual students' needs. The results demonstrate the potential of the system to streamline educational processes, reduce manual effort in exam preparation, and enhance the learning experience through continuous feedback and personalized support. This research highlights the integration of multi-agent systems, automated question classification, and adaptive learning techniques to revolutionize education technology.**

Keywords*— *Artificial Intelligence, Natural Language Processing, Curriculum Optimization, Retrieval-Augmented Generation, Exam Paper Generation, Personalized Learning, Adaptive Learning, Educational Technology, Multi-Agent Systems, Semantic Analysis.

I. Introduction

In modern higher education, students face the challenge of aligning their revision efforts with the actual examination patterns. Misalignments between taught content and exam requirements often lead to inefficient study practices, where students focus on less frequently assessed topics while neglecting crucial ones. This issue is particularly prominent in disciplines with complex subject matter, such as Database Management Systems (DMS), where students must process vast amounts of information from various sources. Traditional exam preparation methods, including manual analysis of past papers and lecture notes, are often time-consuming, inconsistent, and not scalable for large datasets. Thus, there is a growing need for data-driven solutions that can optimize curriculum delivery, improve learning outcomes, and enhance study efficiency.

To address these challenges, we propose a novel AI-driven framework for curriculum optimization, combining multiple advanced technologies such as Natural Language Processing (NLP), machine learning, and data analytics. The system leverages AI to analyze historical exam papers, lecture content, and continuous assessments, automating the generation of aligned model exam papers, personalized study plans, and adaptive feedback for students. This unified system aims to reduce manual effort, enhance exam preparation, and provide a more personalized learning experience.

The framework integrates a multi-agent approach, where specialized agents collaborate to automate content generation and feedback processes. These agents include a Blueprint Analyst for extracting exam structure patterns, a Content Researcher to retrieve relevant lecture content, a Question Writer to generate model exam questions, and a Quality Critic to ensure that the content adheres to academic standards. Additionally, the system employs Retrieval-Augmented Generation (RAG) techniques to dynamically generate short notes and feedback, focusing on frequently tested topics and ensuring that the content is both relevant and aligned with historical assessment trends.

A key innovation of this system is its ability to personalize learning. By incorporating semantic analysis and adaptive learning techniques, the system tailors study plans to individual students' needs, helping them focus on weak areas and prioritize topics based on exam relevance. This system also provides real-time feedback and continuous assessment guidance, ensuring that students are consistently improving their understanding and performance.

Ultimately, this research aims to bridge the gap between curriculum delivery and exam preparation, enhancing both student engagement and academic performance. By automating and personalizing the learning process, the proposed system offers a scalable, efficient, and adaptive solution to the challenges faced by students in modern education. The following sections provide an in-depth

discussion of the system's design, implementation, and its potential impact on education technology.

II. Literature Review

The integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into educational systems has significantly reshaped the optimization of teaching and learning processes. AI holds immense potential in improving assessment preparation by automating various aspects of exam and curriculum analysis. Despite these advancements, a gap remains in effectively aligning unstructured lecture materials with the structured format of exams, particularly for technical courses like Database Management Systems (DBMS)[1].

A. Automatic Question Generation (AQG)

A substantial body of research has focused on Automatic Question Generation (AQG) to enhance educational systems. Early AQG systems were primarily rule-based and template-driven, ensuring grammatical correctness but often falling short of scalability and content depth [2]. With the advent of transformer-based models such as BERT and GPT, more contextually relevant questions have been generated. However, these systems often overlook critical structural elements of exams, such as section divisions, mark distributions, and recurring question formats [3]. Recent advancements have incorporated retrieval-based techniques, such as Retrieval-Augmented Generation (RAG), to ground the generated questions in real-world data [4]. However, few studies have integrated these technologies into domain-specific frameworks, such as exam-focused curriculum alignment [5].

B. Curriculum Alignment and Educational Analytics

Aligning curriculum with assessments has been a longstanding challenge addressed by various educational analytics techniques. AI and NLP offer substantial improvements over traditional methods, which are often inconsistent and labor-intensive. Studies by Romero and Ventura (2020) highlight the use of data-driven techniques like frequency analysis (e.g., TF-IDF) and semantic clustering to identify and prioritize topics for educational content [6]. These methods enable better alignment of syllabus materials with assessments, ensuring that the most frequently tested and conceptually important topics are prioritized in study aids [7]. However, these techniques have largely been used in isolation, without integrating them into real-time exam preparation tools such as model exam paper generation systems [8].

C. Text Summarization and Exam Preparation

Summarization techniques play a critical role in transforming large amounts of lecture content into concise, digestible formats. Early extractive methods, such as TF-IDF, were commonly employed to summarize educational content [9]. However, more recent approaches have shifted toward abstractive summarization, utilizing transformer models like T5 and BART to generate human-like summaries that are not only factually accurate but also contextually relevant[10]. These advanced models offer the potential to produce study materials aligned with exam structures, providing students with tailored notes based on the most relevant content extracted from lecture slides and past exam papers. However, the full potential of these models in exam paper generation, which integrates topic relevance, frequency, and difficulty, remains underexplored [11][12].

D. Semantic Matching and Question Difficulty Prediction

Another key component of exam preparation is understanding the difficulty level of potential exam questions. Semantic matching techniques, particularly those based on embeddings like Sentence-BERT, have been used to match exam questions with relevant lecture materials [13]. These models help identify which sections of the syllabus are most closely aligned with specific types of exam questions. Moreover, the use of AI for predicting the difficulty of exam questions has gained momentum in recent years. Several studies have employed machine learning models, such as fine-tuned transformers, to predict the cognitive difficulty of questions based on the language used in the question and its alignment with frameworks like Bloom's Taxonomy [14][15]. However, these models still struggle to provide real-time, actionable feedback to students on how to improve their study strategies or focus on the most relevant topics based on past exam trends [16].

E. Integrated Learning Systems and Exam Paper Generation

Intelligent tutoring systems (ITS) have made strides in adaptive learning, focusing on personalizing content for students based on their performance and learning needs. ITS frameworks that generate personalized feedback and guide students through learning activities have proven effective in improving student outcomes [17]. However, there is a lack of integrated systems that combine personalized study plans, practice question recommendations, and real-time assessment preparation tools within a unified platform. Current systems primarily focus on either generating static quizzes or summarizing lecture content without adapting to individual student needs or real-time trends in exam patterns [18][19].

III. Methodology

This research proposes an AI-powered framework designed to enhance university students' learning experience by automating exam paper generation, creating aligned short notes, providing personalized study plans, and offering continuous feedback. The system combines advanced AI techniques, Natural Language Processing (NLP), machine learning, and data-driven methodologies, structured through a multi-agent architecture where specialized agents perform specific tasks under the guidance of a central orchestrator.

The system leverages historical exam data, lecture content, and semantic analysis to create a personalized, dynamic learning experience. It automates exam preparation and adapts to individual student needs, providing feedback that supports their academic development.

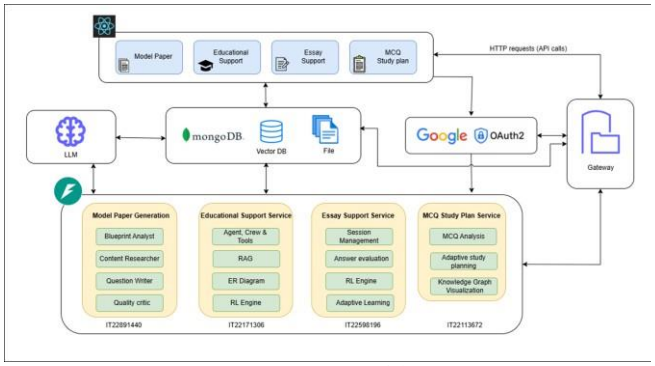


Figure 1:Overview of the system

A. Model paper generation and aligned short notes

The backend of the system is powered by FastAPI, with React for the user interface. The process begins by extracting relevant content from past exam papers and lecture slides, ensuring that the system has the required resources to generate exam papers. The Blueprint Analyst Agent plays a pivotal role by ensuring the alignment of exam topics with the curriculum, validating the question slots, target marks, and structure of the generated exam paper. The Content Researcher Agent retrieves lecture materials using SentenceTransformers and FAISS for efficient content search, ensuring the questions are based on accurate and relevant information.

The Question Writer Agent uses predefined templates to draft the exam questions, utilizing content from the researcher agent. The drafted questions are reviewed by the Quality Critic Agent, which ensures they meet academic standards and are relevant to the intended exam structure. The orchestrator ensures the smooth coordination of agents, handling tasks such as blueprint validation, topic adjustments, and slot management. This orchestrated workflow guarantees that exam papers are generated efficiently, with alignment to the required academic content.

For short notes generation, the system follows a separate pathway. After identifying the relevant question, it retrieves associated content from the lecture slides via the Content Researcher and summarizes it using a large language model (LLM). This ensures that the short notes are curriculum-aligned and concise, providing a focused study aid for students.

B. Study material generation and continuous assessment guidance

The system features several critical functions aimed at enhancing the student learning experience: summarization, continuous assessment guidance, flashcard generation, and ER diagram creation. The Summarization function uses NLP techniques to process and retrieve relevant lecture content stored in the FAISS Vector Database. The content is then summarized by a GPT-3-based model, offering concise, key concept-driven summaries for students. Additionally, the system includes a reinforcement learning mechanism that refines summaries based on student feedback, ensuring continuous improvement.

For continuous assessment (CA) guidance, the system empowers students by offering contextual support and recommendations as they work on their assignments. The CA Guidance Agent extracts questions from submitted PDF assessments and retrieves relevant materials from lecture slides or external resources like academic papers or video tutorials. The system then creates personalized resource recommendations, assisting students in solving problems while also managing deadlines with Google Calendar integration.

Component	Description
Agents	Autonomous entities responsible for specific tasks such as summarizing content, providing guidance, generating flashcards, and assisting with ER diagram creation. Each agent interacts with the student to provide tailored support.
Crew	Oversees the agents to ensure their activities align with the learning goals of the students. The crew coordinates and optimizes agent functions, ensuring the system operates smoothly and efficiently.
Tools	Support resources that assist agents in performing their tasks. Tools handle content retrieval, processing, and feedback collection, enabling real-time information delivery and resource recommendations for students.

Table 1

Flashcard generation is another vital function, supporting active recall by generating topic-specific flashcards based on retrieved content. The system uses Retrieval-Augmented Generation (RAG) to generate flashcards across all levels of Bloom's Taxonomy, enabling students to study at different cognitive levels. User feedback is incorporated into the flashcard generation process, refining the content for future sessions.

In addition to these, the ER diagram generation tool helps students visualize relationships between entities by automatically generating dynamic Entity-Relationship (ER) diagrams using Graphviz. This feature aids in understanding complex concepts that require visual representations, such as database modeling.

C. Essay based learning support system and feedback driven improvement

The Intelligent Essay-Based Adaptive Learning System is designed to help students practice essay-type questions. The workflow starts with extracting essay-type questions from PDFs (including scanned materials) using OCR and text preprocessing techniques. Once validated, questions are classified by difficulty using Bloom's Taxonomy, allowing the system to present questions that align with students' cognitive abilities. This classification is powered by machine learning models, which also ensure that question diversity and relevance are maintained.

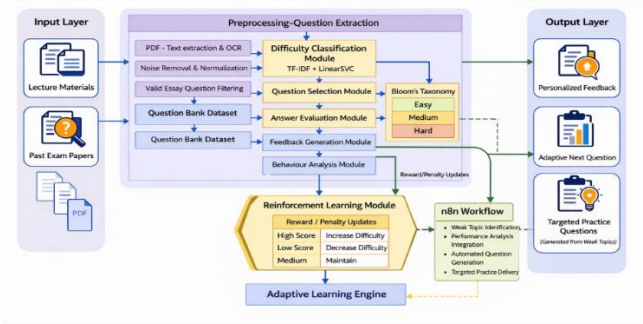


Figure 2:Overview of the system

The Automated Answer Evaluation system uses large language models (LLMs) to evaluate student responses based on content relevance, coherence, and conceptual coverage. Feedback is then provided, highlighting strengths and weaknesses while offering targeted improvement suggestions. The system adapts question difficulty dynamically using a reinforcement learning policy, which adjusts based on student performance—rewarding high scores with increased difficulty, and vice versa.

To complement essay-based practice, the system integrates workflow-based automation to identify weak topics in real-time. By analyzing student performance, the system identifies areas where students struggle the most and generates additional practice questions on these weak topics, ensuring that practice sessions remain focused and aligned with student needs.

D. Personalized study plan generation

The personalized study plan generation component is a core feature, helping students plan their revision efficiently based on their strengths, weaknesses, and available study time. The system begins by semantically understanding lecture notes and historical exam data. Using Sentence-BERT embeddings, lecture topics and MCQs are encoded into high-dimensional vectors, enabling more accurate topic-level mapping.

The system also analyzes past exam papers to determine the importance of various topics, identifying high-frequency topics and computing their relevance based on historical exam trends. These importance scores guide the system in prioritizing topics for the personalized study plan. Additionally, the Graph-based Retrieval-Augmented Generation (Graph RAG) mechanism models relationships between topics, ensuring that study sessions remain contextually relevant and aligned with the exam syllabus.

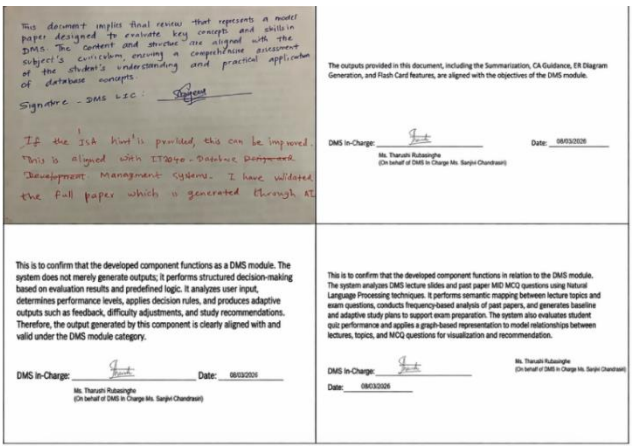
Once the study topics are prioritized, the system dynamically generates a personalized study plan that allocates study hours proportionally to each topic based on its importance and the student's available time. The study plan adapts as the student progresses, ensuring efficient and focused revision. MCQs and practice questions are recommended alongside each study topic, reinforcing the exam-relevant content and enabling continuous assessment.

RESEARCH FINDING & RESULTS

A. Validation by Module Lecturer in Charge

The system was validated by the module lecturer in charge to ensure that the generated model papers, study materials, and feedback mechanisms aligned with the course curriculum and met academic standards.

The validation process involved reviewing the generated content for accuracy, relevance, and alignment with intended learning outcomes. Feedback from the lecturer confirmed that the system could generate exam papers and study resources that accurately reflect the content and structure of previous exams, while maintaining academic rigor. This validation ensured that the system's output was suitable for supporting both educators and students in preparing for assessments.


This document implies final review that represents a model paper designed to evaluate key concepts and skills in DMS. The content and structure are aligned with the subject's curriculum, ensuring a comprehensive assessment of the student's understanding and practical application of database concepts.

Signature - DMS LIC: Student

If the JSA hint is provided, this can be improved. This is aligned with IT2040 - Database Management Development Management System. I have validated the full paper which is generated through AI.

The outputs provided in this document, including the Summarization, CA Guidance, ER Diagram Generation, and Flash Card features, are aligned with the objectives of the DMS module.

DMS In-Charge: Date: 08/03/2026

Ms. Tharushi Rubasinghe
(On behalf of DMS In Charge Ms. Sanjini Chandrasiri)

This is to confirm that the developed component functions as a DMS module. The system does not merely generate outputs; it performs structured decision-making based on evaluation results and predefined logic. It analyzes user input, determines performance levels, applies decision rules, and produces adaptive outputs such as feedback, difficulty adjustments, and study recommendations. Therefore, the output generated by this component is clearly aligned with and valid under the DMS module category.

DMS In-Charge: Date: 08/03/2026

Ms. Tharushi Rubasinghe
(On behalf of DMS In Charge Ms. Sanjini Chandrasiri)

This is to confirm that the developed component functions in relation to the DMS module. The system analyzes DMS lecture slides and past paper MID MCQ questions using Natural Language Processing techniques. It performs semantic mapping between lecture topics and exam questions, conducts frequency-based analysis of past papers, and generates baseline and adaptive study plans to support exam preparation. The system also evaluates student quiz performance and applies a graph-based representation to model relationships between lectures, topics, and MCQ questions for visualization and recommendation.

DMS In-Charge: Date: 08/03/2026

Ms. Tharushi Rubasinghe
(On behalf of DMS In Charge Ms. Sanjini Chandrasiri)

Figure 3:Validation by Module Lecture in Charge

B. Key Findings

1. Alignment with Historical Exam Patterns: The system successfully generated model exam papers that were highly aligned with historical exam structures, ensuring that the questions reflected the typical mark distributions, topics, and difficulty levels of previous exams. This helped simulate real exam conditions for students, aiding in better preparation.

2. Effective Study Material Generation: The system produced concise and relevant study materials, including short notes, flashcards, and summaries, based on high-frequency topics. These resources were highly valued by students for their clarity and focus on key concepts that were likely to be tested in exams.

3. Personalized Learning and Feedback: The integration of personalized feedback and difficulty predictions enabled the system to provide tailored study recommendations, helping students focus on areas where they struggled most. The system's ability to dynamically adjust content based on student performance significantly enhanced learning outcomes.

4. Adaptive Study Plans: The system generated personalized study schedules, helping students manage their time effectively in preparation for exams. These plans were based on factors such as study hours, exam dates, and topic relevance, which optimized the student's revision process.

5. Use of Semantic and Contextual Analysis: The system's use of NLP techniques, including semantic classification and context-aware question generation, enabled it to map lecture topics to relevant exam questions and topics. This ensured that the generated study materials were directly aligned with the content students were learning in their courses.

IV. CONCLUSION & FUTURE WORK

In conclusion, the AI-driven framework developed in this research proves to be an effective tool for automating exam paper generation, personalized study material creation, and providing real-time feedback to students. The system successfully simulates real exam conditions, helps students focus on high-priority topics, and provides tailored learning paths that enhance the learning experience.

Despite some limitations, the positive feedback from students and lecturers demonstrates the system's potential to revolutionize exam preparation and learning across various academic disciplines. Future improvements will focus on expanding the system's scope, optimizing performance, and adding features to further personalize the student experience.

The findings suggest that this AI-powered system can play a critical role in improving educational efficiency, supporting students' learning journeys, and enhancing overall academic performance.

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